

Tropical Proof Systems, Open Problem 4(b) – easy

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1 Preliminaries

A proof system P polynomially simulates a proof system Q if there is a polynomial p such that every Q -refutation of size s has a P -refutation of the same formula of size at most $p(s)$.

Let $\mathbb{Q}_\infty = \mathbb{Q} \cup \{\infty\}$ with $a \oplus b = \min\{a, b\}$ and $a \odot b = a + b$. For every Boolean variable x , introduce tropical variables $x, \bar{x}$ and the equations

$$x \odot \bar{x} = 1, \quad x \oplus \bar{x} = 0.$$

A clause $\ell_1 \vee \dots \vee \ell_k$ is encoded by $1 \leq \ell_1 \odot \dots \odot \ell_k$.

An MP-NSR refutation is an algebraic combination

$$\bigoplus_j t_j \odot f_j \leq \bigoplus_j t_j \odot g_j,$$

where $f_j \leq g_j$ ranges over the encoded clauses, both directions of the Boolean equations, the axiom $0 \leq 0$, and inequalities $g \leq \infty$ for right-hand sides g of the preceding input inequalities. It is a refutation if every monomial on the left has a strictly larger coefficient than the same monomial on the right and the right-hand side has a finite constant term.

A Res(LP) line is a finite disjunction of integer affine inequalities. It starts from the usual linear encodings of clauses and the Boolean axioms

$$x \geq 0, \quad 1 - x \geq 0, \quad (x - 1 \geq 0) \vee (-x \geq 0),$$

and uses

$$\frac{(f_1 \geq 0) \vee \Gamma \quad (f_2 \geq 0) \vee \Gamma}{(\alpha_1 f_1 + \alpha_2 f_2 \geq 0) \vee \Gamma}, \quad \frac{\Gamma}{\Gamma \vee (f \geq 0)},$$

where α_1, α_2 are positive integers. A refutation derives $-1 \geq 0$. A Res(LP) refutation is *treelike* if every derived line is used as a premise at most once. Proof size is the total binary encoding length.

2 Problem formulation

Does treelike Res(LP) polynomially simulate MP-NSR? Namely, is there a polynomial p such that every MP-NSR refutation of size s can be converted into a treelike Res(LP) refutation of size at most

$$p(s)?$$